

Motion in 1-D

Concept of Reference frame $\rightarrow$

- A reference frame is a platform from where a physical phenomenon, such as Motion, is being observed.
- Reference ~~are~~ ~~of~~ frame are of 2 types:
 - (i) Inertial Ref. Frame (non-accelerated)
 - (ii) Non-Inertial Ref. Frame (accelerated)

State of Rest & Motion $\rightarrow$

Rest - When a body does not change its position w.r.t. surroundings & time then it is said to be in the state of Rest.

Motion - When a body changes its position w.r.t. surroundings & time then it is state of Motion.

Distance and Displacement (Relative position vector)

Distance - Actual path length covered by a body is called as Distance.

SI $\rightarrow$ m (meter) CGS $\rightarrow$ cm (centimeter)

Displacement - Change in position vector of a particle going from initial position to final position is called Displacement @ Relative position vector.

Denoted by $\rightarrow \boxed{\Delta \vec{r} = \vec{r}_f - \vec{r}_i}$

Displacement $\leq$ Distance

$$0 \leq \frac{|\text{Displacement}|}{\text{Distance}} \leq 1$$

Average Speed and Average Velocity →

$$\text{Avg. speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\text{Avg. velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

For 1-d we have

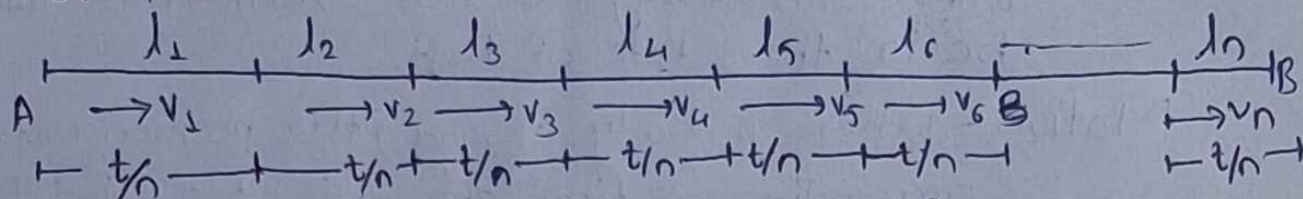
$$V_{av} = \frac{x_f - x_i}{\Delta t}$$

for 3-d we have

$$\vec{V}_{av} = \frac{\vec{r}_f - \vec{r}_i}{\Delta t}$$

concept of Average speed →

(i) for interval divided in equal time parts →

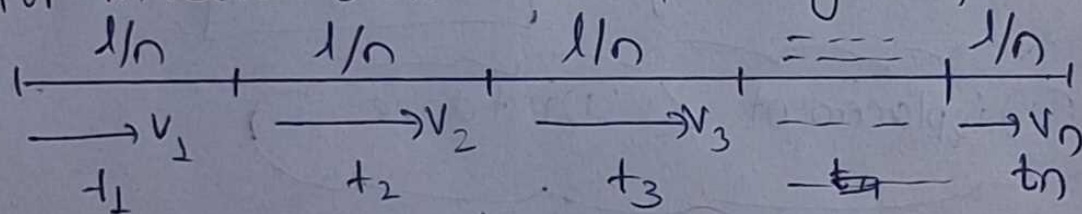


for a body taking 't' time to go from A to B and has velocity v_1 for 1st t/n time, v_2 for next t/n time. & v_n for last t/n time then,

$$V_{av} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n}$$

conclusion → for this type of motion avg. speed is actual average of all velocities.

(ii) for interval divided in equal length parts →



$$\therefore \text{avg. speed} = \frac{1}{\frac{1}{nv_1} + \frac{1}{nv_2} + \dots + \frac{1}{nv_n}} = \frac{1}{\frac{1}{n} \left[\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3} + \dots + \frac{1}{v_n} \right]}$$

$$\therefore \text{avg. speed} = \frac{n}{\frac{1}{v_1} + \frac{1}{v_2} + \dots + \frac{1}{v_n}}$$

$$\therefore \frac{1}{v_{av.}} = \frac{\frac{1}{v_1} + \frac{1}{v_2} + \dots + \frac{1}{v_n}}{n}$$

conclusion: for this type of motion where body travels equal distance for $v_1, v_2, \dots, v_n$ velocities avg. velocity or speed or Inverse is equal to avg. of Reciprocal of velocities

Note $\rightarrow$

- ① when a particle moves with v_1 speed for half the time & ~~v_2~~ with v_2 speed for remaining half time then,

$$v_{av} = \frac{v_1 + v_2}{2}$$

- ② when a particle moves with v_1 speed for half the distance & with v_2 for remaining half distance then,

$$v_{av.} = \frac{2v_1 v_2}{v_1 + v_2}$$

Instantaneous velocity & speed $\rightarrow$

$$\text{Instantaneous speed} = \frac{dx}{dt} = v_{inst.}$$

$$\vec{v}_{inst} = \frac{d\vec{x}}{dt}$$

Acceleration $\rightarrow$
- Rate of change of velocity.-

$$\vec{a}_{\text{avg}} = \frac{\vec{v}_f - \vec{v}_i}{t_2 - t_1} \quad \vec{a}$$

instantaneous accelⁿ $\rightarrow$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{v d\vec{v}}{ds}$$

Conceptual Notes $\rightarrow$

(a) In 3-dimension $\rightarrow$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

(b) $\left| \frac{d\vec{v}}{dt} \right| \rightarrow$ magnitude of accelⁿ.

$\frac{d|\vec{v}|}{dt} \rightarrow$ rate of change of speed
(tangential acceleration, a component of total accelⁿ)

(c) in case of circular motion (uniform)

~~dx~~ speed remains constant

- while velocity changes continuously

$$\therefore \frac{d|\vec{v}|}{dt} = 0 \quad \text{but} \quad \left| \frac{d\vec{v}}{dt} \right| \neq 0$$

(d) it is not possible to have

$$\left| \frac{d\vec{v}}{dt} \right| = 0 \quad \& \quad \frac{d|\vec{v}|}{dt} \neq 0$$

$\rightarrow$ Tangential accelⁿ

$\rightarrow$ Total accelⁿ

Let for an body moving in straight line, with uniform accelⁿ $\rightarrow$ Tangential accelⁿ

- For Uniformly accelerated motion $\rightarrow$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

&

$$S_n = u + \frac{a}{2}(2n-1)$$

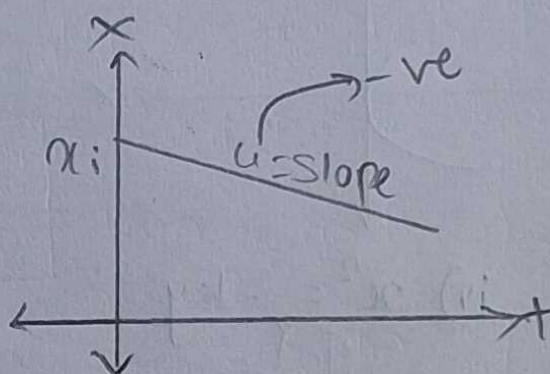
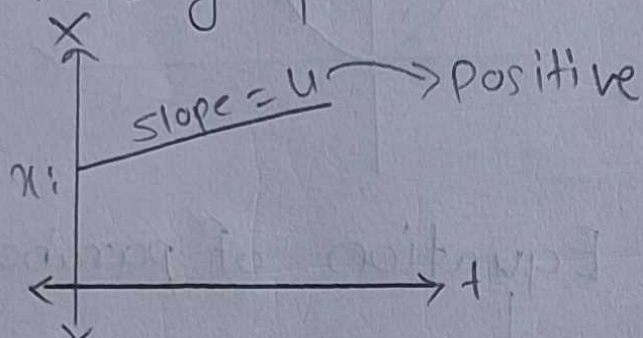
- For Variable acceleration $\rightarrow$

$$a = \frac{d\vec{v}}{dt} = \frac{\vec{v}d\vec{v}}{d\vec{s}}$$

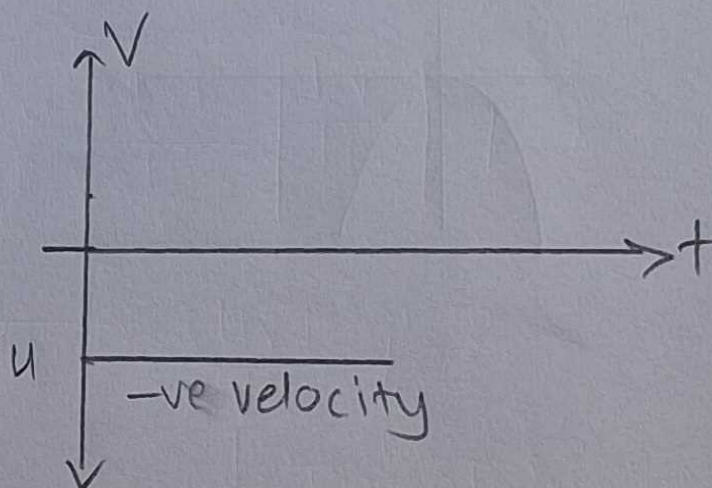
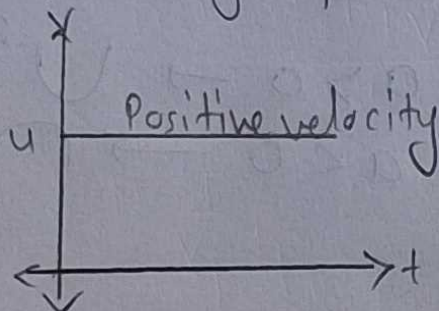
$$v = \frac{d\vec{s}}{dt}$$

Graphs of Motion with Uniform Velocity

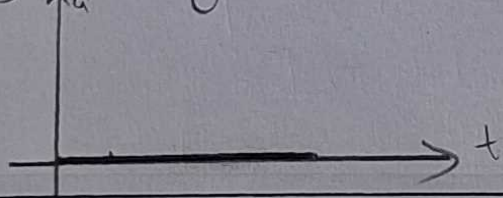
i) $x-t$ graphs $\rightarrow$



ii) $v-t$ graphs $\rightarrow$



iii) $a-t$ graph $\rightarrow$



Graph Identification →

(a) Equation & graphs of line

$$y = mx + c$$

point where
graph cuts the
Y-axis

↪ slope of graph

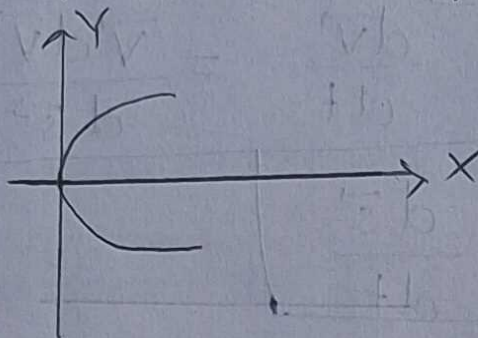
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zero

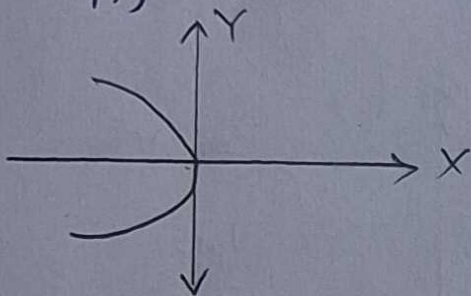
+ve

(b) Parabolic curve - Equation & Graphs.

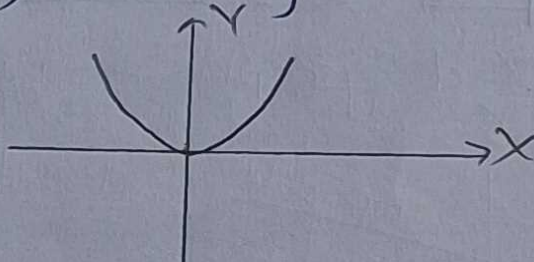
i) $y^2 = kx$



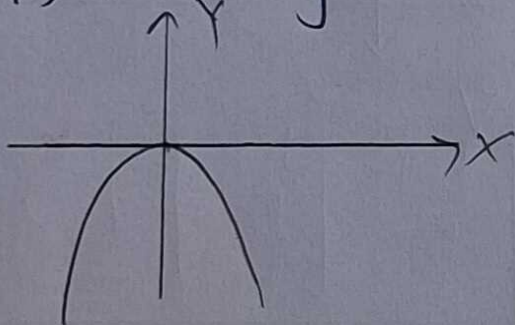
ii) $y^2 = -kx$



iii) $x^2 = ky$



iv) $x^2 = -ky$



Equation of parabola

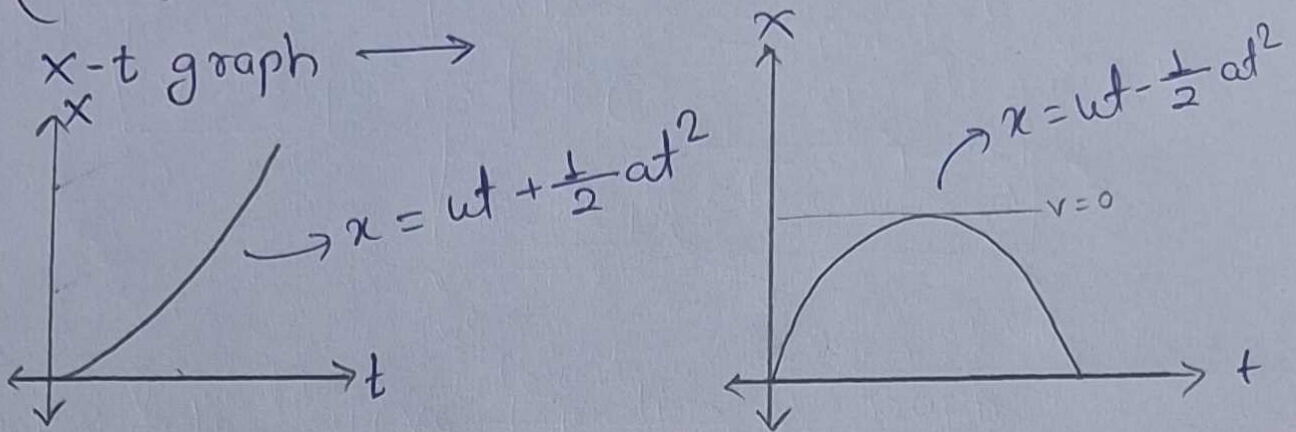
↪ $y = ax^2 + bx + c$

$a > 0 \rightarrow \cup$ {graph

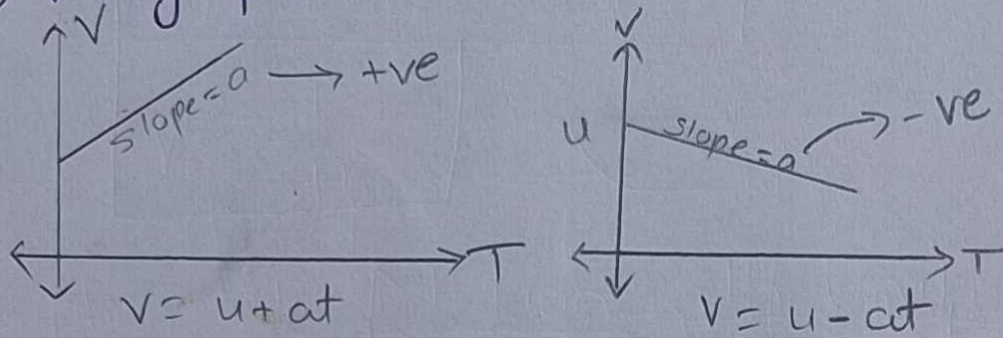
$a < 0 \rightarrow \cap$

Graphs of uniformly accelerated motion $\rightarrow$ ($a \neq 0$)

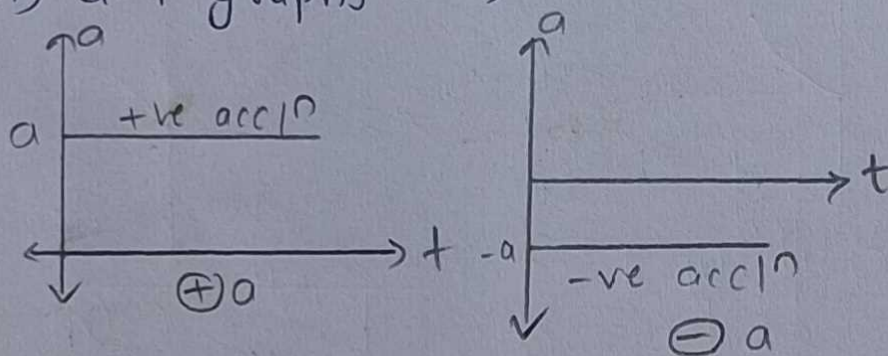
i) $x-t$ graph $\rightarrow$



ii) $v-t$ graphs $\rightarrow$



iii) $a-t$ graphs $\rightarrow$



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